

Tanvir Ahmed Khan

AI Researcher | Computer Vision | Multimodal Learning | 3D Gaussian Splatting | Speech-to-Avatar Systems

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Research Interests

Deep Learning; Computer Vision and Video Generation; Multimodal Models and NLP; Speech Synthesis and Audio Processing; Data-Driven Dynamical Systems (SDEs, Flow Matching, Numerical Methods); World Models and Cognitive AI; 3D Gaussian Splatting.

Education

North South University, Department of Electrical and Computer Engineering - Bachelor of Science in Computer Science and Engineering (2021 - 2025)

- CGPA: 3.80/4.00 | Summa Cum Laude
- Relevant Coursework: Deep Learning, Pattern Recognition, NLP, Numerical Analysis

Rajuk Uttara Model College - Higher Secondary Certificate (HSC), GPA: 5.00/5.00 (2018 - 2020)

Publications

- [Accepted] T. A. Khan et al. "When a Nation Speaks: Machine Learning and NLP in People's Sentiment Analysis During Bangladesh's 2024 Mass Uprising." IEEE International Conference on Computer and Information Technology (ICCIT), 2025.
- [Accepted] T. A. Khan et al. "Ensemble and Temporal Feature-Based Framework for Rainfall Classification in Bangladesh." PLOS ONE, 2025.
- [Under Review] T. A. Khan et al. "CAT-GS: Calibrated Adaptive Gating for Balanced and Robust Multimodal Learning." Neurocomputing, 2026.

Research Experience

Machine Intelligence Lab (MILab), North South University - Research Assistant (Mar 2025 - Present)

Advisors: Dr. Nabeel Mohammed and Dr. Shafin Rahman

- Engineered distributed multi-GPU inference pipelines for real-time talking avatars with sub-4s latency.
- Integrated NVIDIA Audio2Face with Three.js; decoded ARKit blend shapes via linear optimization with high accuracy.
- Trained custom UNet and DiT architectures for high-fidelity 512 x 512 lip-sync mouth refinement.
- Proposed spatio-temporal geometric metrics for evaluating idle facial movements in generative talking faces.
- Developing containerized LLM + TTS conversational agents for Jetson Nano and DGX Spark deployment.
- Designing a unified real-time architecture integrating a speech-to-speech LM with 3D Gaussian Splatting for end-to-end audio-driven talking-head synthesis.

Selected Projects

- Automated Grading System with LLMs - Full-stack grading system (FastAPI + MERN) with RAG pipeline for context-aware evaluation of student submissions.
- DynApex: Dynamic CPU/GPU Parallelization - C++ parallelization tool with data dependency analysis and runtime profiling for dynamic CUDA/OpenMP dispatch.
- Unsupervised Saliency Detection via Knowledge Distillation - Knowledge distillation enabling compact architectures to achieve competitive saliency detection without ground-truth supervision.
- WreckItAll: Physics-Based Game - Physics-based destruction game in Unreal Engine 5.

Industry Experience

Tuctuc, Spain (Remote) - Junior Software Engineer (Jan 2022 - June 2026)

- 3+ years of industry experience maintaining and extending Odoo-based ERP systems, developing React Native mobile apps, and managing GCP cloud infrastructure for scalable deployments.

Technical Skills

- Deep Learning: PyTorch, Transformers, Diffusion Models, GANs, VAEs, Knowledge Distillation, Multi-GPU Distributive Training, NeRF, 3DGS.
- Vision-Language: VLM, World Model, CLIP, LipSync.
- Retrieval and Memory: FAISS, Knowledge Graphs, SQL, RAG Systems.
- NLP: Large Language Models, Sentiment Analysis, Topic Modeling.
- Audio Processing: Mel-Spectrogram Analysis, TTS, NVIDIA Audio2Face.
- Programming: Python, C++, Golang, JavaScript (MERN).
- Tools and Platforms: Git, Docker, GCP, FastAPI, Runpod, Linux.
- Languages: English (Fluent), Bangla (Native).

Awards and Honors

- Academic Scholarship, North South University - 2023.
- Preliminary Qualifier, ICPC Asia Dhaka Regional - 2021.
- Second Prize, Battle of the Bands, NDC Bangladesh - 2019.